DIGITAL FINANCE (OPEN ELECTIVE)

AI & Sentiment Analysis in Investing

How smart computers read market news and social posts to make better investment decisions.

01 / THE BIG IDEA

Numbers Tell the Past, Words Tell the Future

Stock prices usually react to breaking news and company talk before charts show it.

1. Numbers & Charts

Old prices, quarterly profits, and balance sheet tables.

SHOWS WHAT ALREADY HAPPENED

2. Everyday Words

CEO interviews, news headlines, and customer reviews.

SHOWS WHAT IS COMING NEXT

3. AI Sentiment Tool

Rates the text: is it happy (+1), neutral (0), or sad (-1)?

TURNS WORDS INTO SCORES

02 / THE SIMPLE 4-STEP PROCESS

How AI Reads Financial News

From reading raw text on the internet to making a trading choice.

STEP 1

Collect Text

Download live news, tweets, and company reports automatically.

STEP 2

Read & Rate

AI reads each sentence and checks if the tone is good or bad.

STEP 3

Combine Scores

Calculate today's overall mood score for each stock.

STEP 4

Take Action

Buy if sentiment is positive; sell or avoid if negative.

EXAMPLE 01 / REAL-LIFE TEXT

What Does AI See in Company News?

Here is how an AI model (like FinBERT) reads two real statements about a company.

GOOD NEWS EXAMPLE (CEO STATEMENT)

"Our sales grew by 35% this quarter, and orders for our new AI chips are at record highs."

AI Verdict: Happy / Positive (+0.9) • **Signal:** Good time to buy

BAD NEWS EXAMPLE (FINANCE REPORT)

"Store customer visits fell by 10%, and factory supply costs are rising fast."

AI Verdict: Unhappy / Negative (-0.8) • **Signal:** Cut back or sell

Why Computers Do This

Super Fast: Reads 10,000 articles in under 1 second.

No Biases: Does not get tired, greedy, or scared.

Catches Nuance: Knows "growth slowing" is bad even if word "growth" is present.

03 / DAILY PORTFOLIO ROUTINE

How the AI Manages Your Money

A simple continuous loop that keeps an investment portfolio safe and growing.

1

Set Goals

Decide: High growth or safe & steady?

2

Scan Mood

Read the latest news for all companies.

3

Rank Stocks

Put the happiest, strongest stocks on top.

4

Adjust Shares

Buy more winners, reduce struggling stocks.

5

Repeat Daily

Check tomorrow to stay ahead of market shifts.

EXAMPLE 02 / SIMPLE PORTFOLIO EXAMPLE

How $10,000 Gets Rebalanced by AI

See how an AI shifts money when good or bad news hits specific companies.

Company	What the News Said	Before News	After AI Adjusts	Simple Reason
Apple	Very Positive (+0.8)	$2,500 (25%)	**$3,000 (30%)**	Strong holiday iPhone sales
Tesla	Negative (-0.6)	$2,000 (20%)	**$1,400 (14%)**	Price cuts and factory delays
Microsoft	Positive (+0.7)	$3,000 (30%)	**$3,400 (34%)**	Rapid cloud AI growth
Cash Bank	Neutral (Safe)	$2,500 (25%)	**$2,200 (22%)**	Used small cash buffer to buy winners

04 / BIG ADVANTAGES

Why Humans Need AI Help

Machines solve the two biggest weaknesses of human investors: speed and emotions.

Lightning Speed

Reads thousands of stories the instant they go live.

Zero Panic

No fear of missing out and no panic selling at dips.

Watches Everything

Monitors small companies human analysts miss.

Auto-Fixing

Updates your portfolio smoothly as situations evolve.

05 / WATCH OUT!

The Traps of AI Sentiment

AI is powerful, but trusting it blindly can lead to big mistakes.

Old News Lag

By the time news is posted, the price often moved.

Fake Bots & Spam

Fake social accounts create hype to pump dead stocks.

Over-Learning

AI memorizes past patterns that might not work today.

Trading Fees

Buying and selling too often burns money in fees.

EXAMPLE 03 / FRESHNESS MATTERS

How Fast Does News Become Stale?

Different types of news have different lifespans before everyone else knows them.

1. Social Media Posts & Tweets

Expires in: ~10 to 15 Minutes

Very short lifespan; traders react instantly.

2. Quarterly Earnings Call Speeches

Useful for: ~3 to 5 Days

Gives investors several days of good trend signals.

3. Annual Government Audit Filings (10-K)

Useful for: ~1 to 2 Months

Long-term health checks for company stability.

06 / SAFETY FIRST

Rules Before Giving AI Real Money

Three essential safety rules every smart investment system follows.

1. Explain Why

The AI must show which sentences made it buy or sell.

2. Practice Tests

Test the AI on past crashes to see if it survives.

3. Human Boss

A human manager has the final power to stop trades.

EXAMPLE 04 / REAL-LIFE STORY

What Happened During COVID 2020?

A real lesson on why human judgment must guide raw AI systems.

The Mistake (AI Alone)

In March 2020, every headline was terrifying. Raw AI sentiment panicked and sold everything at rock-bottom prices.

Bad Outcome: Missed the huge recovery rally that followed right after.

The Smart Way (AI + Human Guardrails)

Smart systems had built-in pause buttons during extreme panics, preventing panic dumping of quality stocks.

Good Outcome: Kept safe holdings intact and profited as markets bounced back.



THE MAIN LESSON

”Use AI as an Assistant, Not an Autopilot.”

AI is great at reading thousands of articles instantly, but human wisdom and common sense ensure you never crash.

Questions & Open Discussion

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